Sameer Verma

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EDUCATION

University of Cincinnati — **Cincinnati, OH**
BS in Mechanical Engineering, Minor in Mathematics GPA: 3.36/4.0 — *Expected Graduation: May 2029*

- Relevant Courses: Thermodynamics, Solid Mechanics, Dynamics for Engineering, Circuits & Sensing, Material Science, Multivariable Calculus, Differential Equations, Linear Algebra

WORK EXPERIENCE

Gas Dynamics and Propulsion Laboratory — **Cincinnati, OH**
University of Cincinnati — *May 2026 – August 2026 | Full-time | 40 hrs/week*
September 2026 – Present | Part-time | 10 hrs/week

- Engineered a low-cost (~$140) custom strain measurement system for ONR-funded rotating detonation engine research, reducing instrumentation cost by ~98% versus a ~$10,000 conventional system.
- Performed modal FEA in SolidWorks to predict combustor natural frequencies and identify expected structural mode locations in the frequency domain.
- Designed, assembled, and soldered a thermocouple signal-conditioning circuit using an AD8421 instrumentation amplifier, providing ~166x gain (41 µV/°C to 6.8 mV/°C) for combustion diagnostics.
- Built a Python pipeline to convert 1 MHz strain-gauge data to microstrain, apply a Chebyshev Type II low-pass filter, and perform time-frequency spectrogram analysis to interpret combustor structural frequencies and correlate strain response with chamber pressure during post-test review of RDE firing data.
- Diagnosed and mitigated signal-to-noise and EMI issues in the strain measurement chain through iterative improvements to shielding, grounding, amplifier placement, wiring, and high-temperature gauge bonding.

Guerbet — **Cincinnati, OH**
Manufacturing & Quality Intern — *August 2025 – December 2025 | Full-time | 40 hrs/week*

- Tested and validated OptiStar injector assemblies, troubleshooting zeroing calibration and volume discrepancy issues to ensure proper operation.
- Identified and investigated a software-driven volume discrepancy issue using LabVIEW, helping engineers analyze root causes faster by 50% and confirm correct mechanical–software interaction during validation.

LEADERSHIP EXPERIENCE & PROJECTS

Cincinnati Aerospace Society – Liquid Propulsion — **Cincinnati, OH**
Student Engineer, Igniter — *August 2026 – Present*

- Contributing to augmented spark igniter (ASI) design for *Hephaestus*, a 2,500 lbf LOX/CH4 regen-cooled engine.
- Ran NASA CEA/Cantera sweeps of O/F and chamber pressure for the GOX/GCH4 igniter to maximize plume energy while staying within stainless steel wall-temperature limits.
- Ran the igniter material and manufacturing trade study, selecting machined 304L over 3D-printed 316L.

UC Rocketry Club — **Cincinnati, OH**
Student Engineer — *January 2026 – Present*

- Designed a custom battery holder for the rocket's avionics bay using Siemens NX, producing CAD models and manufacturing drawings for flight-hardware integration.
- Used OpenRocket to analyze flight performance, trajectory, and stability throughout design iterations.
- Supported manufacturing, assembly, testing, and launch operations of a two-stage rocket that flew in April 2026 with successful staging.

NASA L'Space NPWEE (Team 18) — **Tempe, AZ (Remote)**
Mechanical Design Lead — *May 2026 – August 2026*

- Co-authored a NASA L'Space proposal investigating plume-surface interactions (PSI) of single-, dual-, and quad-nozzle lunar lander engine configurations on BP-1 regolith simulant.
- Led SolidWorks design of supersonic cold-flow C-D nozzles, swappable acrylic mounting plates, and a 3D-printed PETG regolith tray for a compressed-air plume-impingement test campaign.
- Contributed to a 12-month NASA Phase B–F technology development plan and $6,700 project budget, culminating in a quad chart and New Technology Report (NTR) submission.

Autonomous Multipurpose UAV Drone — **Cincinnati, OH**
Personal Project — *April 2026 – Present*

- Lead mechanical architecture and structural design of a modular UAV airframe in Fusion 360, currently redesigning the fuselage around a removable, serviceable avionics bay.
- Own packaging of propulsion, batteries, electronics, and onboard computing via rapid PETG prototyping with heat-set inserts, balancing structure, cooling, serviceability, and center of gravity.

SKILLS & HONORS

Software: SolidWorks (incl. FEA), Siemens NX, Fusion 360, OpenRocket, NASA CEA
Programming: Python, MATLAB, LabVIEW, Java, VBA
Hands-on: Soldering, Circuit Prototyping, Strain Gauge Installation, 3D Printing / Additive Manufacturing
Honors & Certifications: Eagle Scout (2023); Laser Cutting and Metal Shop A (UC 1819 Innovation Hub)